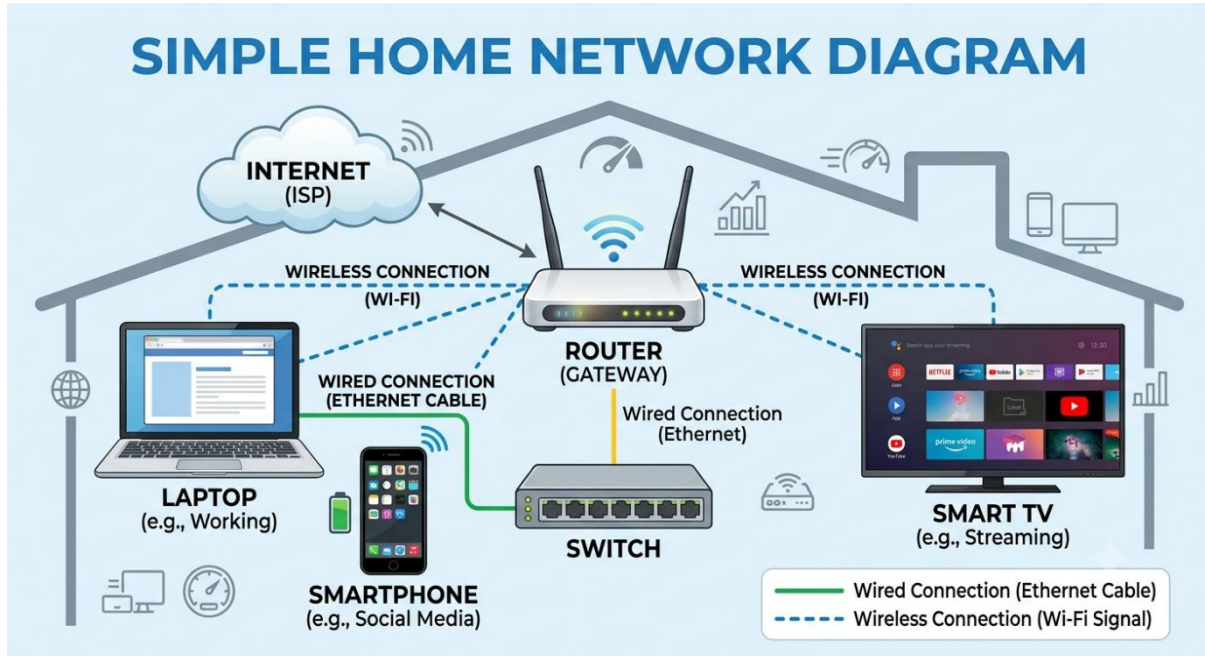


1. Simple Home Network Diagram



Labels:

- **Router:** Connects the home network to the Internet and provides WiFi.
- **Switch:** Expands the number of Ethernet ports.
- **Smartphone:** Connects through WiFi.
- **Laptop:** Connects through Ethernet via the switch.
- **Smart TV:** Connects through WiFi.

2. Comparison Table of Network Devices

Network Device	Main Function	Real-World Use Case
Router	Connects the local network to the Internet and assigns IP addresses.	Connecting home WiFi to browse Google, use WhatsApp, and watch YouTube.
Switch	Connects multiple wired devices in a LAN.	Connecting desktop PCs for LAN gaming or sharing files in an office.
Hub	Broadcasts data to all connected devices.	Used in small or older networks for basic connectivity (rare today).
Access Point (AP)	Extends or creates a WiFi network.	Providing wireless Internet in schools, cafés, or offices for mobile devices.

3. Ordering Food on Zomato

When I order food using the Zomato app on my smartphone, the phone first connects to the Access Point (or the WiFi feature built into the router). The Router receives the request and forwards it to my Internet Service Provider (ISP). If there are multiple wired devices in my home, a Switch connects those devices and passes data within the local network. The router then sends my request over the internet to the Zomato servers. The server processes my order and sends the confirmation back through the ISP, router, access point, and finally to my smartphone, where I can see the order confirmation.

4. Connecting a Gaming PC with Only an Ethernet Port

Option 1: WiFi Extender with Ethernet Port

- Place the WiFi extender halfway between the router and the gaming PC.
- The extender connects wirelessly to the router.
- Connect the gaming PC to the extender using an Ethernet cable.
- The extender acts as a bridge, giving the PC Internet access.

Option 2: Powerline Adapter

- Plug one powerline adapter into a power socket near the router and connect it to the router with an Ethernet cable.
- Plug the second adapter into a power socket near the gaming PC.
- Connect the PC to the second adapter using an Ethernet cable.
- The adapters send network data through the home's electrical wiring, providing a stable wired connection without running a long Ethernet cable.